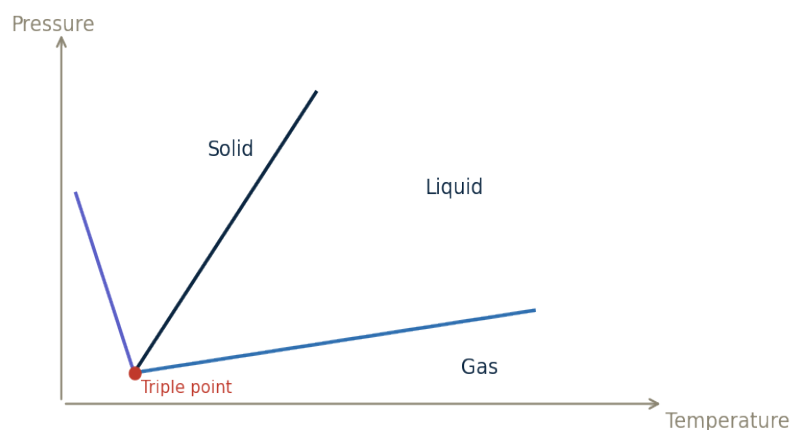


States of Matter

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- The three common states of matter are solid, liquid, and gas, distinguished by the strength of intermolecular forces relative to thermal energy.
- Ideal gas equation: $PV = nRT$, combining Boyle's Law ($P \propto 1/V$ at constant T), Charles's Law ($V \propto T$ at constant P), and Avogadro's Law.
- Real gases deviate from ideal behavior at high pressure and low temperature due to finite molecular volume and intermolecular attractions; corrected by the van der Waals equation.
- Dalton's Law of Partial Pressures: total pressure of a gas mixture = sum of partial pressures of individual gases (assuming no reaction).
- Graham's Law of Diffusion: rate of diffusion of a gas is inversely proportional to the square root of its molar mass.
- A phase diagram shows the conditions of temperature and pressure at which a substance exists as solid, liquid, or gas; the triple point is where all three phases coexist in equilibrium.
- Critical temperature is the temperature above which a gas cannot be liquefied no matter how much pressure is applied.



A phase diagram: regions of solid, liquid, and gas separated by phase-boundary curves.

2. Formula Sheet

Ideal gas equation	$PV = nRT$
Boyle's law	$P_1 V_1 = P_2 V_2$ (constant T)
Charles's law	$V_1/T_1 = V_2/T_2$ (constant P)
Van der Waals equation	$(P + a n^2/V^2)(V - nb) = nRT$
Dalton's law of partial pressures	$P(\text{total}) = P_1 + P_2 + \dots$
Graham's law of diffusion	$r_1/r_2 = \sqrt{M_2/M_1}$
Combined gas law	$P_1 V_1 / T_1 = P_2 V_2 / T_2$
Partial pressure	$P_i = x_i P(\text{total})$

3. Practice Problems (22 questions)

1. A gas occupies 2 L at 1 atm. Find its volume at 2 atm (constant temperature).
2. A gas at 300 K occupies 5 L. Find its volume at 600 K at constant pressure.
3. Find the volume of 2 moles of an ideal gas at 1 atm and 273 K ($R = 0.0821$ L atm/mol K).
4. A mixture of gases has partial pressures 0.4 atm, 0.3 atm, and 0.3 atm. Find the total pressure.
5. Compare the rates of diffusion of hydrogen ($M=2$) and oxygen ($M=32$) using Graham's law.
6. State the postulates of the kinetic molecular theory of gases.
7. A gas cylinder holds 10 L of gas at 5 atm and 300 K. Find the number of moles present ($R = 0.0821$ L atm/mol K).
8. Explain why real gases deviate from ideal gas behavior at high pressure.
9. State Dalton's law of partial pressures and explain the assumption it relies on (no chemical reaction between gases).
10. A gas at 2 atm and 27 degrees C is heated to 127 degrees C at constant volume. Find the new pressure.
11. Define critical temperature and explain its significance for liquefying a gas.
12. Find the pressure exerted by 1 mole of an ideal gas in a 22.4 L container at 273 K ($R = 0.0821$ L atm/mol K).
13. State two ways in which the van der Waals equation corrects the ideal gas equation for real gas behavior.
14. A balloon has volume 3 L at 1 atm and 300 K. Find its volume at 0.5 atm and 300 K.
15. Explain the term 'triple point' shown on a phase diagram.
16. Find the molar mass of a gas if it diffuses 4 times slower than hydrogen ($M=2$) under the same conditions.
17. State Boyle's law and sketch (describe) the shape of a P vs V graph at constant temperature.
18. Explain why gases are more compressible than liquids and solids, based on the kinetic molecular theory.
19. A sample of gas has a volume of 500 mL at STP. Find its volume at 2 atm and 273 K.

20. Explain the difference between an ideal gas and a real gas in terms of intermolecular forces and molecular volume.
21. Find the number of moles of gas in a 5 L container at 2 atm and 400 K ($R = 0.0821 \text{ L atm/mol K}$).
22. State Avogadro's law and explain its molecular-level interpretation.
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Answer Key

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| 1. $V_2 = 1 \text{ L}$ | 2. $V_2 = 10 \text{ L}$ |
| 3. $V = 44.75 \text{ L}$ (approx) | 4. $P(\text{total}) = 1.0 \text{ atm}$ |
| 5. Rate ratio $\text{H}_2:\text{O}_2 = 4:1$ | 6. Conceptual - gas particles are point masses in random motion, elastic collisions, negligible intermolecular forces (ideal case) |
| 7. n approx 2.03 mol | 8. Conceptual - molecules occupy finite volume and attract each other, effects that become significant at high pressure (small volume) |
| 9. Conceptual - total pressure is the sum of partial pressures, assuming gases don't react chemically with each other | 10. P_2 approx 2.67 atm |
| 11. Conceptual - critical temperature is the highest temperature at which a gas can still be liquefied by pressure alone | 12. $P = 1 \text{ atm}$ |
| 13. Conceptual - 'a' term corrects for intermolecular attraction (reduces effective pressure); 'b' term corrects for finite molecular volume (reduces effective volume) | 14. $V_2 = 6 \text{ L}$ |
| 15. Conceptual - the unique condition (T, P) at which solid, liquid, and gas phases coexist in equilibrium | 16. M approx 32 g/mol |
| 17. Conceptual - $PV = \text{constant}$ at fixed T , giving a hyperbolic curve | 18. Conceptual - large intermolecular spacing and weak forces let gas molecules be pushed much closer together under pressure |
| 19. $V_2 = 250 \text{ mL}$ | 20. Conceptual - ideal gas assumes no intermolecular forces and negligible molecular volume; real gases have both, especially significant at high P /low T |
| 21. n approx 0.305 mol | 22. Conceptual - equal volumes of any gas at the same T and P contain equal numbers of molecules |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.